9. Action Group Performance

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9.1 Experiment Objective

This course focuses on learning how to control the robot's action group performance function.

9.2 Experiment Preparation

This course involves the following function from the Muto hexapod robot's Python library:

action(action_id): Action group performance function. The robot has eight built-in action groups, and you can directly call the function to enable the robot's action group performance.

The `action_id` parameter has a value range of [1, 8], representing eight actions, named in order: 1 Stretch, 2 Greet, 3 Retreat in fear, 4 Warm-up squat, 5 Turn in place, 6 Wave to say no, 7 Curl up like a hermit crab, 8 Stride forward.

If you need to end the action performance early, call the `stop()` or `reset()` function.

9.3 Experiment Procedure

Open the JupyterLab client and navigate to the code path:

```
muto/Samples/Control/9.action.ipynb
```

By default, `g_ENABLE_CHINESE` is `False`. If you need to display Chinese, please set `g_ENABLE_CHINESE=True`.

```
# Chinese switch. The default value is English
g_ENABLE_CHINESE = False

Name_widgets = {
    'Stretch': ("Stretch", "Stretch"),
    'Greeting': ("Greeting", "Greet"),
    'Retreat': ("Retreat", "Retreat in fear"),
    'Warm_up': ("Warm_up", "Warm-up squat"),
    'Turn_around': ("Turn_around", "Turn in place"),
    'Say_no': ("Say_no", "Wave to say no"),
    'Crouching': ("Crouching", "Curl up like a hermit crab"),
    'Stride': ("Stride", "Stride forward"),
    'End': ("End", "Finish")
}
```

Click 'Run All Cells', then scroll to the bottom to see the generated controls.

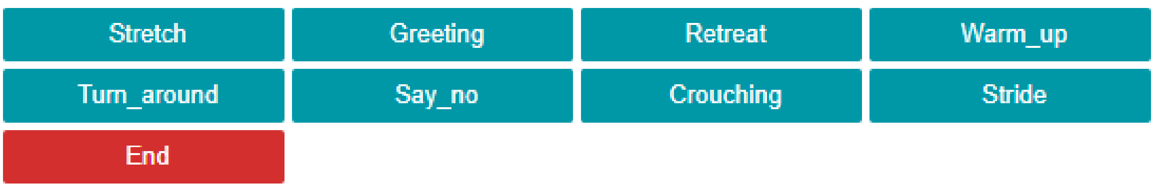
Click the different buttons to perform their corresponding functions.

Each time a button is clicked, the corresponding function is executed. The button event handling is as follows:

```
# Key press event processing
def on_button_clicked(b):
    with output:
        print("Button clicked:", b.description)
    if b.description == Name_widgets['Stretch'][g_ENABLE_CHINESE]:
        g_bot.action(1)
    elif b.description == Name_widgets['Greeting'][g_ENABLE_CHINESE]:
        g_bot.action(2)
    elif b.description == Name_widgets['Retreat'][g_ENABLE_CHINESE]:
        g_bot.action(3)
    elif b.description == Name_widgets['Warm_up'][g_ENABLE_CHINESE]:
        g_bot.action(4)
    elif b.description == Name_widgets['Turn_around'][g_ENABLE_CHINESE]:
        g_bot.action(5)
    elif b.description == Name_widgets['Say_no'][g_ENABLE_CHINESE]:
        g_bot.action(6)
    elif b.description == Name_widgets['Crouching'][g_ENABLE_CHINESE]:
        g_bot.action(7)
    elif b.description == Name_widgets['Stride'][g_ENABLE_CHINESE]:
        g_bot.action(8)
```

9.4 Experiment Summary

In this experiment, JupyterLab controls are used to make the hexapod robot perform its built-in action groups. There are currently eight built-in action groups: 1 Stretch, 2 Greet, 3 Retreat in fear, 4 Warm-up squat, 5 Turn in place, 6 Wave to say no, 7 Curl up like a hermit crab, and 8 Stride forward. Before an action group is completed, clicking another action will have no effect. You must wait for the current action to finish before the next one can be triggered.



To exit the program, press the 'End' button.